Roll No. .....

TME–403

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MANUFACTURING PROCESS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the need of clearance angle and side cutting edge angle ? Explain with neat and clean diagram. (CO1)

OR

(b) What is crater wear and flank wear ? Describe briefly. What are the remedies of avoiding these wears ? (CO1)

2. (a) What would happen if approach angle becomes zero degree ? Discuss in detail with suitable diagram. (CO1)

OR

(b) Show and explain the depth, feed and cutting velocity direction in turning and shaping operation with neat diagram. (CO1)

3. (a) Explain how the total machining time is dependent on feed, r.p.m., and length of workpiece. (CO1)

P. T. O.

OR

(b) Given that back rake angle is 7° and wedge angle is 12°. What would be the end cutting edge angle for the given situation ? (CO1)

4. (a) What is the significance of nose radius on single point cutting tool ? What will happen on the tool life if it is zero degree ? (CO1)

OR

(b) Determine the time of machining during a turning process of a workpiece with diameter 5 mm. Given that length of workpiece is 20 mm, feed is 0.5 mm/rev. and workpiece is rotating at an angular velocity of 10 rad/sec. Number of passes required for the machining is 5. (CO1)

5. (a) Find the shear angle for given data : rake angle is 8°, uncut and cut chip thickness are 2 mm and 3 mm respectively. (CO2)

OR

(b) How can the temperature be minimized during machining a workpiece ? Explain in detail. (CO2)

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TME–404

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MEASUREMENT AND INSTRUMENTATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the basic principle of measurement and primary elements of measuring systems. (CO1)

OR

(b) What are standard ways of measurement ? Explain the importance of standardization. (CO1)

2. (a) Explain the principle and working of the Vernier caliper and Screw gauge. (CO1)

OR

(b) What is the importance of least count ? Calculate the least count of any vernier caliper and screw gauge. (CO1)

3. (a) Discuss several types of tolerances. What is meant by nominal size and tolerance ? (CO1, CO2)

P. T. O.

OR

(b) Explain the importance of toleranceand the major points to be considered while providing tolerance in any system. (CO2)

4. (a) Write a short note on interchangeability and selective assembly. (CO2)

OR

(b) State the Taylor's principle for the design of limit gauge. In addition, also explain the various aspects for deciding the limits on the limit gauges. (CO2)

5. (a) Explain the gauging force. (CO2)

OR

(b) What are the essential consideration in selection of materials for gauges and the common material used for gauges ? (CO2)

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TMS–401

B. TECH. (ME/AME/ROBOTICS) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MANUFACTURING EXECUTION SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain in detail about the concept of 'Tomorrow's Manufacturing'. (CO1)

OR

(b) Discuss in detail about Production systems. Also state the difference between Intermittent and Continuous production systems. (CO1)

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TMS–401

B. TECH. (ME/AME/ROBOTICS) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MANUFACTURING EXECUTION SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain in detail about the concept of 'Tomorrow's Manufacturing'. (CO1)

OR

(b) Discuss in detail about Production systems. Also state the difference between Intermittent and Continuous production systems. (CO1)

2. (a) Distinguish between Mass production systems and Cellular production systems. (CO1)

OR

(b) Discuss in detail about the concept of Production planning horizons. (CO1)

3. (a) Write in detail about the significance of 'Ernerprise Integration'. Also state the steps involved for its implementation in an organisation. (CO1, CO2)

P. T. O.

OR

(b) Discuss in detail about the process of 'Advanced Manufacturing Planning' with suitable diagram. (CO1, CO2)

4. (a) Explain in detail about the concept of 'Material Requirement Planning' along with its primary objectives. (CO3)

OR

(b) Illustrate the difference between short-term and long-term capacity planning strategies. (CO3)

5. (a) Discuss in detail about the significance of 'Enterprise Resource Planning' in an organisation. (CO3)

OR

(b) Explain in detail about the concept of 'Product Lifecycle Management'. Also discuss different phases involved in it. (CO3)

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TME–405

B. TECH. (ME) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

KINEMATICS OF MACHINES

Time : 1½ Hours

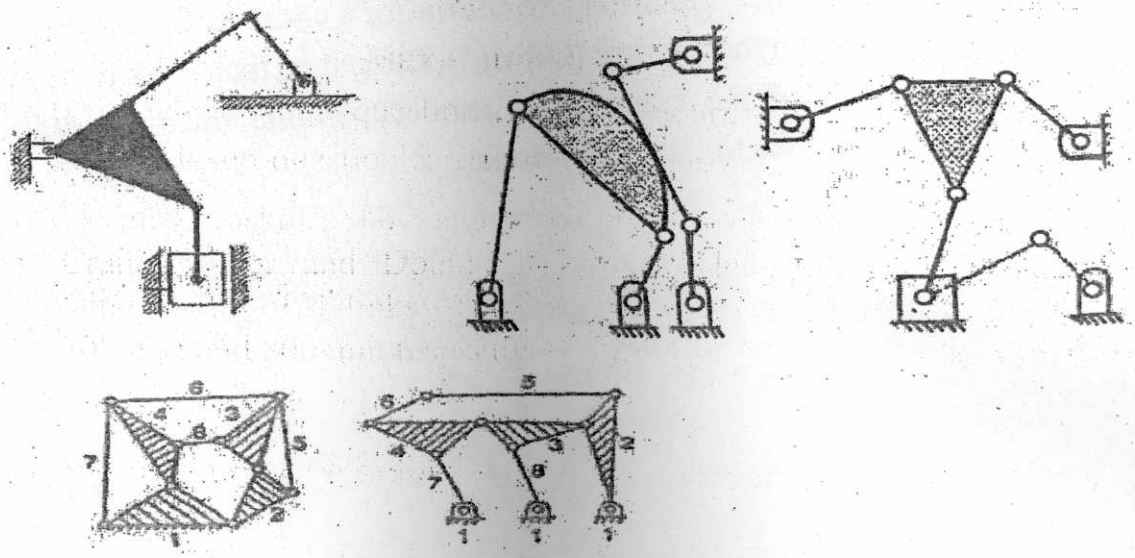
Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define Grashof's law and draw the configuration of Crank-Crank, Crank-Rocker and Double Rocker Mechanism for the links with Dimensions of 4 links of a 4-Bar mechanism as 100 mm, 400 mm, 700 mm and 800 mm respectively. (CO1)

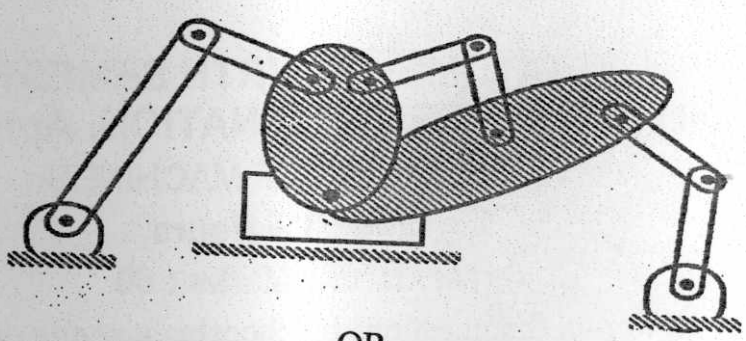
OR

(b) What is the degree of freedom of the mechanisms shown below ? (CO1)



P. T. O.

2. (a) In a kinematic chain with 12 binary links, 8 binary joints, 2 ternary joints, 1 quaternary joint and 2 higher pair, evaluate DOF of the kinematic chain and define its nature. Also, calculate DOF for the following mechanism : (CO1)



OR

(b) Explain and derive the expression for Kutzbach and Grubler's criterion. (CO1)

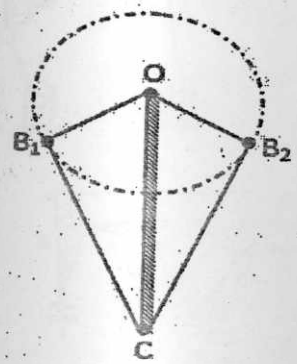
3. (a) Describe the following Mechanism : (CO2)

(i) Elliptical Trammel

(ii) Oldham's Coupling

OR

(b) Write down the names of the different inversions of single slider kinematic chain ? In the figure shown for a crank and slotted lever quick return motion mechanism (QRMM), Length of fixed link is 160 mm and the length of the crank is 40 mm. Determine the time ratio of the working stroke to the return stroke. (CO2)



4. (a) Sketch a pantograph, explain its working and show that it can be used to reproduce to an enlarged scale a given figure. (CO2)

OR

(b) Sketch and describe the Peucellier straight line mechanism indicating clearly the conditions under which the point P on the corners of the rhombus of the mechanism generates a straight line. (CO2)

5. (a) Define Steering Gear mechanism and derive an expression for condition required for correct steering. Also, differentiate between Davis and Ackermann Steering Mechanisms. (CO2)

OR

(b) Define Kinematic pair. What are the different types of kinematic pairs ? Explain them. (CO1)

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TME–406

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MECHANICS OF MATERIALS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

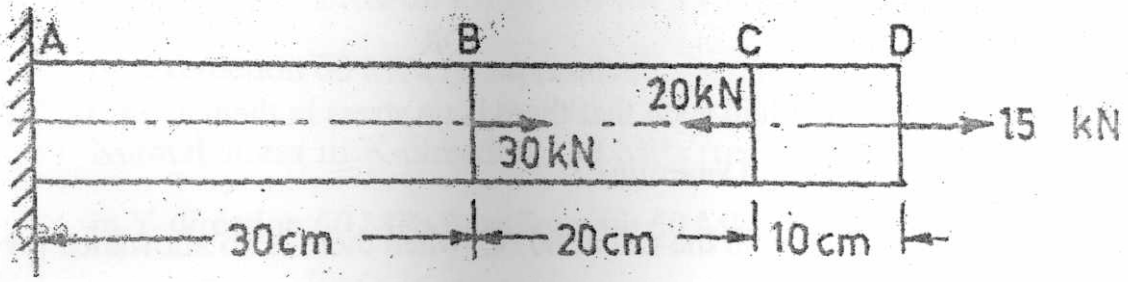
(ii) Each sub-question carries 10 marks.

1. (a) Deduce expression to determine the elongation of a bar of tapering circular section. (CO1)

OR

(b) A steel bar of 30 mm diameter is loaded as shown in Figure. Determine the stress in each portion and the total elongation. E = 210 GPa.

(CO1)

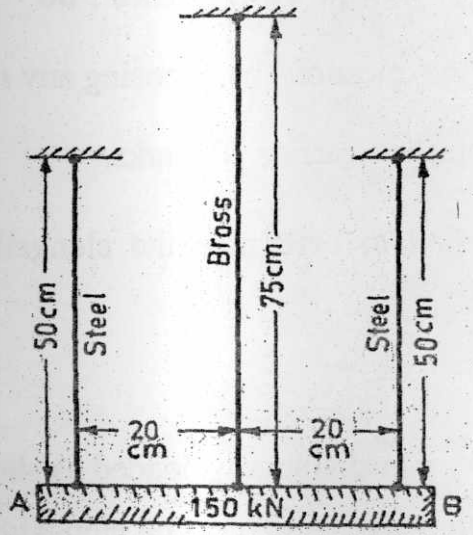


P. T. O.

2. (a) Find the expression for elongation of a bar of rectangular section due to self weight. (CO1)

OR

(b) A rigid bar AB is supported by three bars in the same vertical plane and are equidistant from each other as shown in Figure. The outer bars are made of steel, 2.5 cm diameter and 50 cm long each. The central bar is made of brass and is 75 cm long and 2 cm diameter. Calculate the forces in the bars due to an applied force of 150 kN assuming that AB remains horizontal after the load is applied. Assume $E_{(steel)}$ = 210 GPa and $E_{(brass)}$ = 105 GPa. (CO1)



3. (a) What do you mean by thermal stresses ? Explain. (CO2)

OR

(b) Rails are laid such that there is no stress in them at 24° C. If the rails are 32 m long. Determine : (CO2)

(i) Stress in the rail at 80° C, when there is no allowance for expansion

(ii) Stress in the rail at 80° C, when there is expansion allowance of 8 mm per rail.

(iii) The expansion allowance for no stress in the rail at 80°C.

Take :

Coefficient of linear expansion = $11 \times 10^{(-6)}/°C$,

$$E = 205 \text{ GPa}.$$

4. (a) What do you mean by principal planes and principal stresses ? Derive the expression for principal stresses for a body subjected to direct and shear stresses. (CO2)

OR

(b) Draw Mohr's stress circle indicating the radius and centre point coordinate for given stress conditions : (CO2)

(i) Pure shear (60 MPa)

(ii) Hydrostatic stresses (120 MPa)

(iii) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa(tensile)

(iv) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa (compressive)

(v) Normal stress in X-direction 120 MPa (tensile) and Normal stress in Y-direction 60 MPa (tensile) with 60 MPa shear stress

5. (a) What is Mohr's stress circle ? How is the Mohr's circle constructed ? (CO2)

OR

(b) The stresses on two perpendicular planes through a point are 120 MPa tensile, 80 MPa compressive and 60 MPa shear. Determine the normal and shear stress components on a plane at 60° to that of the 120 MPa stress and also the resultant and its inclination with the normal component on the plane. (CO2)

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TME–408

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

APPLIED THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the air standard Otto cycle and derive an expression for its efficiency. (CO2, CO4)

OR

(b) Explain the various performance parameters of an internal combustion engine. (CO1)

2. (a) The minimum pressure and temperature in an Otto cycle are 100 kPa and 27°C. The amount of heat added to the air per cycle is 1500 kJ/kg.

(i) Determine the pressures and temperatures at all points of the air standard Otto cycle.

(ii) Also calculate the specific work and thermal efficiency of the cycle for a compression ratio of 8 : 1.

Take for air : $C_v = 0.72$ kJ/kg0K, and $\gamma = 1.4$. (CO3, CO6)

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OR

(b) The pressure and temperature of air at the beginning of compression in an Otto cycle is 103 kPa and 27°C, respectively. The heat added per kg of air is 1850 kJ. The compression ratio is 8. Determine :

(i) maximum temperature

(ii) maximum pressure

(iii) thermal efficiency

(CO3, CO6)

3. (a) Write the name of different processes constitute the following air standard cycles and also draw their P-v and T-s diagram : (CO2)

(i) Stirling cycle

(ii) Lenoir cycle

(iii) Atkinson cycle

OR

(b) Compare the efficiency of air standard Otto, Diesel and Dual cycles for :

(i) the same compression ratio and same maximum heat rejection, and

(ii) the same maximum pressure and temperature, and same maximum heat rejection. (CO2, CO4)

4. (a) Explain the effect of pressure ratio on the net work output and efficiency of an ideal Brayton cycle. Also derive the expression of optimum pressure ratio for maximum net work output. (CO3)

OR

(b) Consider an air standard Brayton cycle in which the air enters the compressor at 1 bar and 20°C. The pressure of air leaving the compressor is 3.5 bar and the temperature at turbine inlet is 600°C. Determine per kg of air : (CO3, CO6)

(i) Efficiency of the cycle

(ii) Heat supplied to air

(iii) Net work done

(iv) Heat rejected in the cooler

(v) Temperature of air leaving the turbine.

5. (a) Explain the effect of regeneration and intercooling on Brayton cycle with the help of T-s diagram. (CO4)

OR

(b) What do you mean by jet and rocket propulsion ? Explain the working of ramjet and turbojet engine with the help of a neat sketch.

(CO1, CO2)

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TME–407

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

FLUID MECHANICS AND FLUID MACHINES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) State the Newton's law of viscosity. Explain the importance of viscosity in fluid motion. What is the effect of temperature on viscosity ? (CO1, CO2, CO4)

OR

(b) Determine the intensity of shear of an oil having viscosity = 1 poise. The oil is used for lubricating the clearance between a shaft of diameter 10 cm and its journal bearing. The clearance is 1.5 mm and the shaft rotates at 150 r.p.m. (CO1, CO2, CO4)

2. (a) State and prove Pascal's law. (CO1, CO2, CO4)

OR

(b) A U-tube manometer is used to measure the pressure of water in a pipe line, which is in excess of atmospheric pressure. The right limb of the

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manometer contains mercury and is open to atmosphere The contact between water and mercury is in the left limb. Determine the pressure of water in the main line if the difference in level of mercury in the limbs of U-tube is 10 cm and the free surface of mercury is in level with the center of the pipe. (CO1, CO2, CO4)

3. (a) What is Total Pressure Force and Centre of Pressure ? Determine the force exerted on an Inclined Plane Surface Submerged in Liquid. (CO1, CO2)

OR

(b) Find the magnitude and direction of the resultant force due to water acting on a roller gate of cylindrical form of 4.0 m diameter, when the gate is placed on the dam in such a way that water is just going to spill. Take the length of the gate as 8 m. (CO1, CO2)

4. (a) What is Continuity Equation ? Derive the expression for continuity equation in 3-D. (CO2, CO4)

OR

(b) A rectangular pontoon 10.0 m long. 7 m broad and 2.5 m deep weighs 686.7 kN. It carries on its upper deck an empty boiler of 5.0 m diameter weighing 588.6 kN. The centre of gravity of the boiler and the pontoon are at their respective centres along a vertical line. Find the meta-centric height. Weight density of sea water is 10.104 kN/m^3. (CO2, CO4)

5. (a) What is a Venturimeter ? Derive the expression for rate of flow through a venturimeter. (CO3, CO4)

OR

(b) An orifice meter with orifice diameter 15 cm is inserted in a pipe of 30 cm diameter. The pressure difference measured by a mercury oil differential manometer on the two sides of the orifice meter gives a reading of 50 cm of mercury. Find the rate of flow of oil of sp. gr. 0.9 when the co-efficient of discharge of the orifice meter is 0.64.

(CO3, CO4)